Chapter 3E: Inflammatory Bowel Disease, Irritable Bowel Syndrome, and Functional/Inflammatory Colitis

Original study resource for clinical reasoning, inpatient/outpatient management, and graph-RAG extraction

This chapter consolidates the chronic inflammatory and functional bowel disorders that are commonly confused on internal medicine exams and in clinical practice. The organizing question is not simply whether a patient has diarrhea; it is whether the bowel is inflamed, structurally damaged, infected, ischemic, medication-injured, or functionally hypersensitive. Inflammatory bowel disease (IBD) is a chronic immune-mediated disorder with objective inflammation, while irritable bowel syndrome (IBS) is a disorder of gut-brain interaction without mucosal injury. Many colitis syndromes sit between these categories clinically because they produce diarrhea and abdominal pain but differ in histology, urgency, complications, and treatment.

The source pages supplied for this segment emphasize Crohn disease, ulcerative colitis, classic extraintestinal manifestations, complications, and an abdominal-pain diagnostic framework. The content below uses those topics as a coverage checklist while adding current diagnostic and treatment organization, chronic diarrhea contrast, microscopic colitis, ischemic/infectious mimics, medication-induced disease, and graph-friendly relationship maps.

High-yield numbers to anchor this chapter include: Crohn disease commonly involves the terminal ileum; ulcerative colitis involves the rectum in nearly all classic cases; extraintestinal manifestations occur in a clinically important minority of IBD patients; toxic megacolon is suggested by systemic toxicity with colonic dilation often around 6 cm or more; C. difficile should be considered during flares; fecal calprotectin above roughly 50-100 micrograms/g supports intestinal inflammation rather than purely functional disease; and long-standing extensive colitis increases colorectal cancer risk, making surveillance colonoscopy part of chronic care.

Why this matters clinically

IBD and IBS represent a common diagnostic fork. A 24-year-old with years of crampy pain, urgency, blood, weight loss, aphthous ulcers, and anemia should not be labeled as IBS. A 38-year-old with abdominal pain related to bowel movements, alternating stool form, normal hemoglobin, normal inflammatory markers, and no nocturnal symptoms may have IBS and does not need the same escalation pathway as IBD. The distinction matters because IBD can cause strictures, fistulas, abscesses, toxic megacolon, venous thromboembolism, and colorectal cancer, whereas IBS causes substantial suffering but does not destroy bowel or raise cancer risk.

A second clinical fork is Crohn disease versus ulcerative colitis. Crohn disease behaves like a full-thickness inflammatory process that can narrow, penetrate, and skip. Ulcerative colitis behaves like a continuous superficial colitis that begins in the rectum and spreads proximally. Those anatomic differences explain most exam findings: fistulas and strictures favor Crohn disease; toxic megacolon and primary sclerosing cholangitis classically favor ulcerative colitis; colectomy can be curative for colonic ulcerative colitis but not for Crohn disease because Crohn disease can recur elsewhere in the gastrointestinal tract.

Core illness scripts

Disorder	Typical patient and symptoms	Key objective findings	Clinical danger
Crohn disease	Young adult with chronic RLQ pain, nonbloody diarrhea or mixed diarrhea, weight loss, fever, mouth ulcers, perianal disease, or obstructive episodes.	Skip lesions, transmural inflammation, ileitis, cobblestoning, fistulae, abscesses, strictures, possible noncaseating granulomas.	SBO, abscess, fistula, malabsorption, kidney stones, gallstones, recurrent disease after surgery.

Disorder	Typical patient and symptoms	Key objective findings	Clinical danger
Ulcerative colitis	Young adult with urgency, tenesmus, frequent small-volume stools, hematochezia, cramping, and symptoms beginning at the rectum.	Continuous mucosal inflammation from rectum proximally, crypt abscesses, friability, pseudopolyps; no skip lesions in classic disease.	Severe colitis, toxic megacolon, perforation, hemorrhage, colorectal cancer risk, PSC association.
IBS	Recurrent abdominal pain associated with defecation and change in stool frequency or form, often with bloating; symptoms wax and wane.	Normal colonoscopy when performed, normal inflammatory markers, no anemia, no weight loss, no nocturnal inflammatory diarrhea.	Low mortality but high quality-of-life burden; danger is missing IBD, cancer, infection, celiac disease, or microscopic colitis.
Microscopic colitis	Older adult, often female, with chronic watery nonbloody diarrhea; may be associated with NSAIDs, PPIs, SSRIs, smoking, or autoimmune disease.	Grossly normal colonoscopy but abnormal biopsies showing lymphocytic or collagenous colitis.	Persistent dehydration/weight loss; missed diagnosis if biopsies are not taken from normal-appearing mucosa.
Ischemic colitis	Older patient or vascular-risk patient with acute crampy abdominal pain and hematochezia, often after hypotension or low-flow state.	Segmental colitis, classically watershed areas; CT may show wall thickening; colonoscopy confirms when stable.	Gangrene, perforation, sepsis; can mimic IBD or infection.

Original relationship map: genetic susceptibility, epithelial barrier dysfunction, microbiome shifts, and mucosal immune activation connect to persistent bowel inflammation. Crohn disease expresses that inflammation as transmural, discontinuous injury with strictures and fistulas; ulcerative colitis expresses it as continuous mucosal injury beginning in the rectum. Treat-to-target care links symptoms, biomarkers, imaging, and endoscopic healing rather than symptoms alone.

Pathophysiology: what separates inflammation from functional symptoms

IBD is not just a symptom label. It is a chronic inflammatory state arising from the interaction of genetic susceptibility, epithelial barrier dysfunction, intestinal microbiome changes, and dysregulated mucosal immunity. In Crohn disease, inflammation extends through the bowel wall. Full-thickness involvement permits sinus tracts, fistulas, phlegmon, abscess, and fibrotic narrowing. In ulcerative colitis, inflammation is typically limited to the mucosa and submucosa, so the disease causes bleeding, urgency, mucus, and diffuse friability rather than fistulas through the bowel wall.

IBS, by contrast, is a disorder of visceral hypersensitivity, altered motility, microbiome signaling, and gut-brain interaction. The mucosa is not ulcerated, the bowel wall is not progressively destroyed, and inflammatory biomarkers are generally normal. IBS can follow infectious gastroenteritis, occur with anxiety or stress amplification, and coexist with other functional pain syndromes. The key is that IBS is a positive diagnosis when the symptom pattern fits and red flags are absent; it should not be used as a wastebasket diagnosis for unexplained weight loss, anemia, nocturnal diarrhea, blood, fever, or abnormal inflammatory markers.

Crohn disease

Crohn disease can involve any region from mouth to anus, but the terminal ileum and ileocecal region are classic. The pattern is discontinuous: abnormal segments alternate with normal segments. Because disease is transmural, inflammation can behave in three broad ways: inflammatory disease causing pain and diarrhea; stricturing disease causing obstructive episodes; and penetrating disease causing fistulas, abscesses, and phlegmon. A single patient may move from one phenotype to another over time.

The classic presentation is chronic or recurrent abdominal pain, often in the right lower quadrant, with diarrhea, malaise, low-grade fever, and weight loss. Diarrhea is often nonbloody when disease is small-bowel predominant, but colonic Crohn disease can bleed. Terminal ileal disease may impair bile acid absorption and vitamin B12 absorption. Bile acid malabsorption contributes to watery diarrhea, while chronic B12 deficiency can produce macrocytosis and neurologic symptoms. Fat malabsorption and inflammation can cause weight loss and low albumin.

Complications are strongly tied to transmural disease. Strictures arise from chronic edema, muscular hypertrophy, and fibrosis, producing colicky pain, distention, vomiting, and obstructive episodes. Fistulas can connect bowel to bowel, bladder, vagina, skin, or perianal tissue. Perianal fissures, abscesses, and fistulas are particularly suggestive of Crohn disease rather than ulcerative colitis. Abscess should be suspected when a patient has fever, focal tenderness, leukocytosis, or persistent symptoms despite steroids.

Crohn disease also has metabolic consequences. Ileal inflammation reduces bile acid reabsorption, which can promote cholesterol gallstones. Unabsorbed bile acids entering the colon can worsen diarrhea. Fat malabsorption increases colonic oxalate absorption, predisposing to calcium oxalate kidney stones. Chronic inflammation also increases risk for venous thromboembolism, osteoporosis, anemia of chronic disease, iron deficiency, and malnutrition.

Ulcerative colitis

Ulcerative colitis is a chronic inflammatory disease of the colon that classically begins in the rectum and extends proximally in a continuous pattern. Because rectal involvement is central, symptoms often include urgency, tenesmus, frequent small-volume stools, mucus, and hematochezia. Proctitis causes rectal bleeding and urgency; left-sided colitis adds cramping and diarrhea; extensive colitis can produce fever, tachycardia, anemia, hypoalbuminemia, and systemic toxicity.

The inflammation in ulcerative colitis is usually mucosal rather than transmural. That explains why fistulas and deep strictures are not typical. However, widespread mucosal inflammation can be severe enough to cause toxic megacolon, hemorrhage, perforation, and systemic inflammatory collapse. Severe colitis is a hospital-level disease when there are frequent bloody stools, tachycardia, fever, anemia, elevated inflammatory markers, dehydration, or radiographic colonic dilation.

Ulcerative colitis is also linked to primary sclerosing cholangitis (PSC), a cholestatic bile duct disease that may progress independently of bowel symptoms. PSC increases the risk of cholangiocarcinoma and colorectal cancer. Colectomy can remove the diseased colon and cure colonic ulcerative colitis, but it does not necessarily eliminate PSC or all extraintestinal manifestations.

Feature	Crohn disease	Ulcerative colitis
Distribution	Mouth-to-anus potential; terminal ileum and ileocecal region are common; skip lesions.	Colon-limited in classic disease; rectum first; continuous proximal spread.
Depth	Transmural inflammation.	Mucosa and submucosa predominant.
Stool pattern	Often nonbloody diarrhea when small bowel predominant; can be bloody with colitis.	Bloody diarrhea, urgency, mucus, tenesmus are common.
Gross/endoscopic clues	Cobblestoning, aphthous ulcers, strictures, fistulas, patchy lesions.	Diffuse friability, superficial ulceration, pseudopolyps, continuous disease.
Histologic clue	Noncaseating granulomas can occur but are not required.	Crypt abscesses and chronic mucosal architectural distortion.
Complications	Fistulas, abscesses, strictures, SBO, malabsorption, perianal disease.	Toxic megacolon, severe hemorrhage, colorectal cancer risk, PSC.
Surgery	Treats complications but is not curative; recurrence after resection is common.	Colectomy can be curative for colonic disease.

Extraintestinal manifestations

IBD is systemic. Some extraintestinal manifestations parallel bowel activity, while others may progress independently. Peripheral arthritis, erythema nodosum, episcleritis, and aphthous ulcers often flare with intestinal inflammation. Ankylosing spondylitis, sacroiliitis, uveitis, and PSC may not track with bowel symptoms. This distinction matters: treating the colon may improve erythema nodosum but may not resolve PSC or axial spondyloarthritis.

System	Manifestations	Clinical notes
Eyes	Episcleritis, uveitis	Episcleritis often parallels bowel activity; uveitis can threaten vision and may not parallel gut symptoms.
Skin	Erythema nodosum, pyoderma gangrenosum, aphthous ulcers	Erythema nodosum is tender nodules, often on shins; pyoderma gangrenosum is ulcerative and can worsen with trauma.
Joints	Peripheral arthritis, migratory mono/oligoarthritis, sacroiliitis, ankylosing spondylitis	Peripheral arthritis often parallels bowel activity; axial disease can be independent.
Hepatobiliary	Primary sclerosing cholangitis, gallstones	PSC is classically associated with UC; gallstones are linked to ileal Crohn disease and bile acid malabsorption.
Renal/metabolic	Calcium oxalate nephrolithiasis, osteoporosis	Oxalate stones can follow fat malabsorption in Crohn disease; steroids and inflammation worsen bone loss.
Vascular	DVT, PE, arterial events	Active IBD is a hypercoagulable state; hospitalized flares require VTE prophylaxis unless contraindicated.

Diagnostic approach to suspected IBD

The first step is to decide whether the clinical syndrome is inflammatory. Blood, fever, nocturnal diarrhea, weight loss, elevated CRP, iron deficiency anemia, hypoalbuminemia, and high fecal calprotectin all support inflammation. Stool testing is essential because infection can mimic a flare, and C. difficile can occur in patients with IBD even without classic antibiotic exposure. A new diagnosis of IBD should not be made solely by symptoms; it requires endoscopic, histologic, and often imaging evidence.

Colonoscopy with ileoscopy and biopsies is the key test for most stable patients. Biopsies should be taken even from areas that look normal because chronic microscopic changes help distinguish IBD from infection or ischemia. If Crohn disease is suspected beyond the reach of colonoscopy, CT enterography or MR enterography can evaluate small-bowel inflammation, strictures, abscesses, and fistulas. Capsule endoscopy can visualize mucosal disease but should be avoided when a stricture is suspected unless patency is confirmed.

Fecal calprotectin is useful because neutrophil activity in the intestinal lumen raises the value in inflammatory disease. A value below a low threshold makes active IBD less likely in many outpatient settings, while a high value supports inflammation but does not identify the cause. Infection, NSAID enteropathy, colorectal cancer, and other inflammatory processes can also elevate calprotectin. Therefore, calprotectin is a triage and monitoring tool, not a standalone diagnosis.

Step	Branch	Action
Chronic diarrhea or recurrent abdominal pain	Red flags present: blood, weight loss, fever, anemia, nocturnal symptoms	Order stool infection tests, CBC, CRP/fecal calprotectin, then colonoscopy with biopsies and ileoscopy when appropriate.
Chronic diarrhea or recurrent abdominal pain	No red flags; stable labs; pain linked to defecation	Consider IBS, diet triggers, celiac testing when appropriate, and limited evaluation.
Suspected Crohn complication	Fever, focal pain, obstruction, perianal disease	Use CT/MR enterography or pelvic MRI depending on phenotype.

IBD severity and emergency recognition

IBD severity is not determined only by stool count. A patient with six stools daily, tachycardia, fever, abdominal tenderness, anemia, and hypoalbuminemia is sicker than a patient with eight stools daily but normal vital signs and labs. Severe disease can progress to toxic megacolon, perforation, or sepsis. Hospital-level features include high fever, persistent tachycardia, orthostasis, severe anemia, acute kidney injury, peritoneal signs, inability to tolerate oral intake, suspected abscess, severe dehydration, or colonic dilation.

Toxic megacolon is a key life-threatening complication, especially in ulcerative colitis but also in infectious colitis and Crohn colitis. It represents severe colonic inflammation with loss of neuromuscular tone, dilation, systemic toxicity, and risk of perforation. Clinical clues include fever, tachycardia, leukocytosis, anemia, abdominal distention, tenderness, and radiographic colonic dilation. Management includes bowel rest, IV fluids, correction of electrolytes, broad evaluation for infection, IV corticosteroids when IBD is the driver, early surgical consultation, and avoidance of antidiarrheals, opioids, anticholinergics, and colonoscopy insufflation when dilation is severe.

Clinical problem	Suggestive findings	Immediate priorities
Severe UC flare	Frequent bloody stools, fever, tachycardia, anemia, high CRP, dehydration.	Hospitalize if severe; stool C. difficile; IV fluids; IV corticosteroids; VTE prophylaxis; rescue therapy or surgery if nonresponse.
Crohn abscess	Fever, focal pain, leukocytosis, persistent symptoms despite steroids, palpable mass.	CT abdomen/pelvis or enterography; antibiotics; drainage when feasible; avoid escalating immunosuppression until infection controlled.
Fibrostenotic Crohn obstruction	Crampy pain, distention, vomiting, prior Crohn surgery, narrowed segment on imaging.	Bowel rest, fluids, NG decompression if needed, imaging; distinguish inflammatory edema from fixed fibrosis; surgery/endoscopic dilation for fixed strictures.
Toxic megacolon	Systemic toxicity plus colonic dilation, abdominal distention, severe colitis.	ICU-level monitoring when unstable, IV steroids if IBD, antibiotics if concern for infection/perforation, urgent surgical involvement.

Management principles in IBD

Modern IBD management is based on induction of remission, maintenance of remission, prevention of complications, and objective monitoring. Symptom improvement alone is not enough because inflammation can persist silently. The treat-to-target strategy uses symptoms, biomarkers such as CRP and fecal calprotectin, endoscopic healing, and imaging when needed. Medication choice depends on disease location, severity, risk factors, prior therapy, complications, pregnancy considerations, infection risk, and patient preferences.

Corticosteroids are induction drugs, not maintenance drugs. They can rapidly reduce inflammation but do not maintain durable remission safely and cause infection risk, hyperglycemia, hypertension, osteoporosis, mood changes, cataracts, and adrenal suppression. A patient who repeatedly needs prednisone is not controlled; they need steroid-sparing therapy. Steroid-sparing options include immunomodulators, biologics, and targeted small molecules depending on disease type and severity.

Before biologic or advanced immunosuppression, clinicians screen for latent tuberculosis, hepatitis B, vaccination status, and infection risk. Live vaccines are generally avoided once significant immunosuppression has begun. Hospitalized IBD patients should be considered for pharmacologic VTE prophylaxis unless bleeding risk clearly outweighs benefit, because active inflammation increases thrombotic risk.

Medication class	Role	Important cautions
5-ASA / mesalamine	Most useful in mild to moderate ulcerative colitis, especially colonic/rectal disease. Rectal formulations help proctitis and left-sided disease.	Not a core therapy for moderate to severe Crohn disease. Monitor renal function in selected patients.
Corticosteroids	Short-term induction for flares; budesonide has more local activity in ileocecal Crohn or mild UC formulations depending on product.	Do not use as maintenance. Watch for infection, bone loss, hyperglycemia, psychiatric effects.
Thiopurines / methotrexate	Steroid-sparing maintenance or combination therapy in selected patients.	Delayed onset; monitor CBC/LFTs; infection and malignancy considerations; methotrexate is teratogenic.

Medication class	Role	Important cautions
Anti-TNF agents	Moderate to severe Crohn or UC, fistulizing Crohn, steroid-dependent disease.	Screen for TB/hepatitis B; infection risk; immunogenicity; may combine with immunomodulator in selected patients.
Anti-integrin / anti-interleukin agents	Biologic options for moderate to severe disease, especially when anti-TNF is not ideal or fails.	Choice depends on phenotype, prior therapy, comorbidities, speed needed, and safety profile.
JAK inhibitors / S1P modulators and other small molecules	Selected moderate to severe UC and some Crohn contexts depending on agent and approval.	Thrombosis, infection, lipid, cardiovascular, pregnancy, and monitoring issues vary by agent.

Crohn disease treatment by phenotype

Mild ileocecal inflammatory Crohn disease may respond to budesonide for induction, but recurrent or higher-risk disease requires a maintenance plan. Higher-risk Crohn features include young age at diagnosis, extensive small-bowel involvement, deep ulcers, perianal disease, fistula, abscess, stricturing disease, prior surgery, smoking, and steroid dependence. Moderate to severe inflammatory disease usually requires biologic or advanced therapy rather than repeated steroid bursts.

Penetrating Crohn disease requires both medical and procedural thinking. An abscess should be drained when feasible before intensifying immunosuppression. Perianal fistulizing disease often requires pelvic MRI or endoscopic ultrasound, surgical seton placement, antibiotics for sepsis control, and biologic therapy. Stricturing Crohn disease requires the clinician to decide whether narrowing is inflammatory and potentially steroid/biologic responsive, or fibrotic and more likely to need dilation or surgery.

Surgery in Crohn disease is not failure; it is often necessary for obstruction, abscess, perforation, dysplasia/cancer, or refractory localized disease. The key counseling point is that surgery is not curative. Recurrence can occur at the anastomosis or elsewhere, so postoperative risk assessment and surveillance are essential.

Ulcerative colitis treatment by extent and severity

Ulcerative proctitis is treated differently from pancolitis. Rectal mesalamine is highly effective for proctitis because it delivers therapy directly to the inflamed rectum. Left-sided disease often uses both oral and rectal mesalamine. More extensive or moderate disease may require oral steroids for induction if 5-ASA fails, but maintenance should be steroid-free. Moderate to severe UC is usually treated with biologic or targeted small-molecule therapy chosen according to severity, comorbidities, prior exposure, and safety issues.

Acute severe ulcerative colitis is an inpatient emergency. Patients need infection testing, including C. difficile, fluid and electrolyte correction, DVT prophylaxis, abdominal imaging if distended, and IV corticosteroids unless contraindicated. Lack of improvement within several days should trigger rescue therapy such as infliximab or cyclosporine in appropriate settings, or colectomy if deterioration, perforation, uncontrolled bleeding, or refractory disease occurs.

Colectomy is curative for colonic UC but still carries functional consequences. It is indicated for perforation, cancer/dysplasia not manageable endoscopically, toxic megacolon, uncontrolled hemorrhage, fulminant steroid-refractory disease, and chronic medically refractory disease. Patients may undergo ileal pouch-anal anastomosis, end ileostomy, or staged procedures depending on acuity and surgical factors.

IBS and disorders of gut-brain interaction

IBS is defined by recurrent abdominal pain associated with defecation and changes in stool frequency or stool form. The stool pattern can be diarrhea-predominant (IBS-D), constipation-predominant (IBS-C), mixed, or unclassified. Bloating is common. Pain often improves or changes after a bowel movement. Symptoms can be triggered by stress, meals, FODMAP-rich foods, infection, sleep disruption, or visceral hypersensitivity.

The most important step is to check for alarm features. IBS should not cause GI bleeding, iron deficiency anemia, persistent fever, nocturnal diarrhea, unintentional weight loss, a new onset at older age, progressive symptoms, a family history of colon cancer or IBD with concerning features, or abnormal objective inflammatory markers. When red flags are absent, a limited evaluation is often sufficient. Celiac serology is reasonable in diarrhea-predominant IBS-like symptoms. CRP or fecal calprotectin can help separate IBS from IBD in uncertain cases.

Treatment is symptom-pattern based. Education and reassurance are therapeutic because patients often fear cancer or progressive disease. Diet strategies include lactose reduction when lactose intolerance is likely and a structured low-FODMAP trial when bloating and meal-related symptoms dominate. IBS-D options include loperamide for diarrhea frequency, bile acid binders in selected patients, rifaximin in selected cases, and neuromodulators when pain is prominent. IBS-C options include soluble fiber, osmotic laxatives, secretagogues, and pelvic floor evaluation if evacuation difficulty dominates. Tricyclic antidepressants can reduce pain and diarrhea but may worsen constipation; SSRIs may help pain and comorbid anxiety in selected patients.

Feature	IBS	IBD
Inflammation	Absent or not the driver; normal mucosa and normal inflammatory biomarkers expected.	Objective intestinal inflammation with endoscopic, histologic, imaging, or biomarker evidence.
Bleeding	Not expected. Blood requires another diagnosis.	Common in UC; possible in Crohn colitis.
Nocturnal symptoms	Uncommon; symptoms often linked to waking, meals, stress.	Can occur, especially with active inflammation.
Weight loss/anemia	Should prompt evaluation for organic disease.	May occur from inflammation, bleeding, malabsorption, poor intake.
Cancer/structural complications	No increased CRC, fistula, or stricture risk from IBS itself.	CRC risk with long-standing colitis; Crohn can cause fistula/stricture/abscess.
Treatment frame	Diet, bowel habit control, neuromodulation, gut-brain education.	Anti-inflammatory/immunosuppressive therapy, monitoring, complication prevention.

Microscopic colitis and other inflammatory mimics

Microscopic colitis is a common missed cause of chronic watery diarrhea. Colonoscopy may look normal, so the diagnosis requires biopsies. Two histologic patterns are classically described: lymphocytic colitis and collagenous colitis. Symptoms are usually nonbloody watery diarrhea, urgency, nocturnal stools, and sometimes weight loss. Associations include older age, female sex, smoking, autoimmune disease, NSAIDs, proton pump inhibitors, SSRIs, and other medications. Budesonide is commonly used for induction when symptoms are significant, while medication review and antidiarrheals may help milder disease.

Infectious colitis can mimic IBD, especially when fever, blood, and leukocytes are present. C. difficile deserves special attention because it can occur in IBD patients and can precipitate severe flares. Stool testing should be performed in apparent IBD exacerbations with significant diarrhea, especially if hospitalized, recently exposed to antibiotics, immunosuppressed, or severely ill.

Ischemic colitis is often acute, segmental, and pain-predominant, with hematochezia following crampy abdominal pain. It is common in older adults and low-flow states, but it can also occur with vasoconstrictive drugs, thrombophilia, endurance exercise, or vascular disease. It differs from chronic IBD by timing, vascular risk context, segmental distribution, and rapid onset. Severe ischemia with peritoneal signs, gangrene, or persistent sepsis requires surgical evaluation.

Medication-induced colitis should be considered when symptoms begin after a new drug. NSAIDs can cause enteropathy and colitis; immune checkpoint inhibitors can produce severe immune-mediated colitis; mycophenolate can mimic IBD; antibiotics can cause C. difficile; and metformin, magnesium, colchicine, and laxatives can cause diarrhea without colitis. A medication timeline is a diagnostic test.

Question	IBS-favoring answer	Inflammatory colitis-favoring answer
Are there alarm features?	No bleeding, anemia, weight loss, fever, or nocturnal diarrhea.	Blood, anemia, fever, weight loss, nocturnal symptoms, or abnormal inflammatory markers.
What do labs/stool markers show?	Normal CBC and low inflammatory markers.	Elevated CRP or fecal calprotectin; leukocytosis; hypoalbuminemia; iron deficiency.
What is the next test?	Positive IBS diagnosis with limited testing if stable.	Stool testing and colonoscopy with biopsies; image if Crohn complication suspected.

Common pitfalls

• Calling chronic diarrhea IBS before asking about nocturnal symptoms, bleeding, weight loss, anemia, fever, family history, and medication exposures.

• Treating repeated IBD flares with repeated steroid bursts without a steroid-sparing maintenance plan.

• Escalating immunosuppression in Crohn disease without excluding abscess or infection when fever and focal pain are present.

• Forgetting that normal-appearing colonoscopy does not exclude microscopic colitis unless biopsies are taken.

• Missing C. difficile in an apparent ulcerative colitis flare.

• Using antidiarrheals or opioids in severe colitis or suspected toxic megacolon.

• Assuming colectomy cures all systemic problems in UC; PSC and some extraintestinal manifestations can persist.

• Forgetting VTE prophylaxis in hospitalized active IBD patients when not contraindicated.

High-yield numeric anchors

Number	Meaning
Fecal calprotectin >50-100 micrograms/g	Supports intestinal inflammation over a purely functional disorder in many outpatient contexts; interpret with pretest probability.
Toxic megacolon: colon often >=6 cm	Radiographic dilation with systemic toxicity is dangerous; avoid antidiarrheals and involve surgery early.
UC rectal involvement	Classic UC begins in the rectum and extends continuously; rectal sparing should raise alternative explanations or treatment effect.
Crohn recurrence after surgery	Surgery is not curative; postoperative recurrence is common and requires surveillance/maintenance planning.
Long-standing colitis	Extensive duration and extent increase colorectal cancer risk; surveillance colonoscopy becomes part of chronic care.
Active IBD hospitalization	Inflammation increases thrombotic risk; use VTE prophylaxis unless contraindicated.

Graph-RAG extraction map

Node	Useful attributes	Important edges
Crohn disease	Transmural, skip lesions, terminal ileum, fistula, stricture, perianal disease, malabsorption, recurrence after surgery.	causes -> SBO; causes -> fistula; causes -> B12 deficiency; associated_with -> kidney stones/gallstones; treated_by -> steroids/biologics/surgery.
Ulcerative colitis	Continuous mucosal colitis, rectal involvement, bloody diarrhea, urgency, toxic megacolon, PSC, colorectal cancer risk.	causes -> hematochezia; associated_with -> PSC; complication -> toxic megacolon; treated_by -> 5-ASA/steroids/biologics/colectomy.

Node	Useful attributes	Important edges
IBS	Functional disorder, pain linked to defecation, altered stool form/frequency, no alarm features, normal inflammatory markers.	mimics -> IBD; treated_by -> diet/loperamide/fiber/neuromodulators; ruled_against_by -> bleeding/anemia/weight loss.
Microscopic colitis	Watery diarrhea, normal colonoscopy appearance, biopsy diagnosis, medication associations.	mimics -> IBS-D; diagnosed_by -> random colon biopsies; treated_by -> budesonide/medication withdrawal.
Toxic megacolon	Systemic toxicity, colonic dilation, severe colitis, perforation risk.	complicates -> UC/infectious colitis/Crohn colitis; treated_by -> bowel rest/IV steroids/surgery consult; avoid -> opioids/antidiarrheals.
Fecal calprotectin	Stool neutrophil marker, inflammatory triage, monitoring tool.	supports -> IBD when elevated; helps_distinguish -> IBS; not_specific_for -> infection/cancer/NSAID injury.

Practice questions

1. A 23-year-old has recurrent RLQ pain, nonbloody diarrhea, weight loss, perianal drainage, and CT evidence of terminal ileitis with a small abscess. What diagnosis is most likely and what management error should be avoided?

Answer: Crohn disease with penetrating complication. The key error is escalating steroids or biologics before addressing abscess/infection. Management requires imaging, antibiotics, drainage when feasible, and then a long-term Crohn plan.

2. A 28-year-old has urgency, tenesmus, frequent small-volume bloody stools, and continuous inflammation from the rectum to the sigmoid colon. What disease pattern is this?

Answer: Ulcerative colitis, likely proctosigmoiditis or left-sided disease depending on extent. Continuous rectal involvement, urgency, tenesmus, and hematochezia favor UC over Crohn disease.

3. A 67-year-old woman has 4 months of watery diarrhea, including nocturnal stools. Colonoscopy looks normal. She takes a PPI and an SSRI. What is the next diagnostic step?

Answer: Colon biopsies to evaluate for microscopic colitis. Normal visual colonoscopy does not exclude microscopic colitis.

4. A patient with known UC presents with fever, tachycardia, abdominal distention, leukocytosis, and colonic dilation. Which therapies should be avoided?

Answer: Avoid opioids, anticholinergics, antidiarrheals, and unnecessary colonoscopic insufflation. This is concerning for toxic megacolon; stabilize, test for infection, give appropriate IV therapy, and involve surgery early.

5. A 34-year-old has recurrent abdominal pain associated with bowel movements and alternating constipation and diarrhea for years. She has no bleeding, weight loss, anemia, nocturnal symptoms, or inflammatory marker elevation. What diagnosis is reasonable and why?

Answer: IBS is reasonable because the symptom pattern fits a disorder of gut-brain interaction and red flags/objective inflammation are absent. Treatment should be symptom-based rather than immunosuppressive.

References and source-use note

This chapter was written as original Medvale educational content. The uploaded GI source pages were used as a coverage checklist for Crohn disease, ulcerative colitis, extraintestinal manifestations, complications, and abdominal-pain framing. External guideline sources consulted for clinical alignment included American College of Gastroenterology IBD guidance and updates, American Gastroenterological Association guidance on moderate-to-severe UC, and standard guideline principles for fecal calprotectin, endoscopic assessment, steroid-sparing maintenance, infection screening, and treat-to-target monitoring.